MALIK JIRASEK

malik.jirasek@student.uni-tuebingen.de · *+49 175 2221257* · *malikjirasek.github.io*

Institute for Theoretical Physics
University of Tübingen, Germany
Auf der Morgenstelle 14
72076 Tübingen

EDUCATION

Oct 2025 – Present
Master of Science in Physics, University of Tübingen, Germany
Thesis: *"Keldysh-field-theory approach to the quantum Fisher information"*
Supervisor: Igor Lesanovsky

Aug 2020 – Jul 2025
Bachelor of Science in Physics, University of Tübingen, Germany
Final grade: 1.04 / 5.0 (German system; 1.0 = excellent)
Thesis: *"Quantum Sensing beyond the Heisenberg Limit with the Boundary Time Crystal"*
Supervisor: Igor Lesanovsky

RESEARCH EXPERIENCE

Nov 2025 – Present
Master of Science Thesis Research, University of Tübingen, Germany
- Developing a Keldysh-field-theory description of the quantum Fisher information in open quantum systems, with a focus on scalable sensor architectures.
- Demonstrating genuine quantum advantage in driven-dissipative many-body sensor architectures.

Aug 2025 – Present
Working Student, Bosch eBike Systems, Germany
- Implementing a Machine Learning pipeline for battery cell aging predictions, using BiLSTM networks and TensorFlow.
- Collaborating with the engineering team to integrate the model into the production workflow.

Oct 2024 – Jul 2025
Bachelor of Science Thesis Research, University of Tübingen, Germany
- Demonstrated sub-Heisenberg scaling of precision in optical-phase estimation using a boundary time crystal as a light source.
- Designed and numerically benchmarked realistic measurement protocols with QuTiP simulations.
- Results incorporated into a manuscript submitted to *Physical Review Letters* (under review).

Aug 2024 – Oct 2024
Research Intern, Humboldt University of Berlin, Germany
- Extended a Fortran-based many-body solver to include high-precision interaction potentials for ultracold-atom tweezer arrays, achieving 10^{-8} relative precision in ground state energies.

PUBLICATIONS

In preparation
Malik Jirasek, Igor Lesanovsky, and Viktoria Noel, "Keldysh-field-theory approach to the quantum Fisher information". *In preparation*.

Preprint / under review
Malik Jirasek, Igor Lesanovsky, and Albert Cabot, "The Boundary Time Crystal as a light source for collectively enhanced sensing", arXiv:2511.23416 [quant-ph], 2025. Accepted for publication in *Physical Review A*.

PRESENTATIONS & CONFERENCES

01 Oct 2026	Accepted for poster at DQUT Conference, Palma de Mallorca
07 Sep 2026	Accepted for poster at QUANT26 Master school, Dresden
06 Mar 2026	Contributed Talk at DPG Spring Meeting SAMOP, Mainz

TEACHING EXPERIENCE

WS 2024/25	Advanced Quantum Mechanics tutor (University of Tübingen)
SS 2024	Quantum Mechanics tutor (University of Tübingen)
WS 2023/24	Analytical Mechanics tutor (University of Tübingen)
SS 2023	Physics 2 tutor (University of Tübingen)
WS 2022/23	Physics 1 tutor (University of Tübingen)
WS 2021/22 – WS 2024/25	Physics Lab tutor (University of Tübingen)

AWARDS & SCHOLARSHIPS

2021 & 2023	Deutschlandstipendium (merit-based; awarded to ~0.5% of enrolled students)

SKILLS & METHODS

Programming	Python (TensorFlow, QuTiP), Fortran, MATLAB (Simulink)
Simulation & Modeling	Lindblad formalism, Keldysh formalism, stochastic calculus
Tools	Linux, Git, LaTeX
Languages	German (native), English (full professional proficiency), French (beginner)

SERVICE & LEADERSHIP

Oct 2026	Co-organizer, Workshop "Nonequilibrium methods in driven-dissipative many-body systems"
2026	Student representative, faculty appointment committee (ML in physics)
2022 – 2025	Treasurer & member, physics student council

REFERENCES

Available on request.